MALMÖ UNIVERSITY

Designing Beyond the Default User

Reflective Interaction Design for Exposing Hidden Assumptions in Digital Systems

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Abstract

Digital systems are often presented as neutral and universal despite being shaped by hidden assumptions regarding users' bodies, behaviours, abilities, and needs. This bachelor thesis in Interaction Design, explores how such assumptions become embedded within interaction design and how they influence perceptions of inclusivity, neutrality, and adaptation in digital systems. Through literature research, interviews, user testing, participatory design, and speculative redesigns of systems such as ChatGPT and Google Maps, the project investigated how hidden assumptions manifest within everyday interaction experiences. The redesigned prototypes introduced features focused on transparency, contextual adaptation, safety, and user control. Interestingly, while the original systems were frequently described as "for everyone" or "neutral," the redesigned versions were often perceived as "more for women" despite being designed to support broader inclusivity and flexibility. This suggested that many assumptions embedded within existing interaction systems may remain invisible precisely because they have become normalized. When redesigns instead attempted to expose, challenge, or adapt beyond these norms, participants became more likely to perceive the systems as targeted toward specific groups rather than universal or neutral. The final outcome of the thesis is a reflective design toolkit aimed at helping interaction designers identify, question, and negotiate hidden assumptions throughout the design process. The toolkit was further explored through the development of a final redesigned ChatGPT prototype informed by participatory testing and reflective design principles. The thesis argues that inclusive interaction design is not solely about adding marginalized users into existing systems, but about critically examining who systems are designed around from the beginning. Through reflective and transparent design approaches, the project explores how interaction design may help reduce the gap between imagined users embedded within systems and the diverse realities of actual human experience.

Keywords: Interaction Design, Reflective Design, Feminist HCI, Research through Design, Hidden Assumptions, Inclusive Design, Adaptive Interfaces

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1 Introduction

"*Humanity is male and man defines woman not in herself, but as relative to him.*"
— de Beauvoir, as cited in Criado Perez (2019, p. 12). The quote raises an important question for interaction design: if what is normalized becomes perceived as universal, then who is today's "default user" actually designed around and can interaction design ever truly be neutral?

Contemporary interaction systems are often designed and perceived as universal, neutral, and equally accessible to all users. However, previous research within Human-Computer Interaction (HCI), Feminist HCI, and critical technology studies has repeatedly demonstrated that many digital systems instead reflect historically normalized assumptions regarding bodies, behaviours, cognition, technological confidence, and social experiences (Bardzell, 2010; Criado Perez, 2019; Cirillo et al., 2020). While these assumptions frequently remain invisible within everyday interaction, they may nevertheless shape who interaction systems primarily support, who is expected to adapt, and which experiences become treated as "normal" within digital environments.

This thesis explores how speculative redesign and reflective interaction design approaches may help expose hidden assumptions embedded within contemporary digital systems. While gender functions as an important lens throughout the project, the thesis does not primarily aim to investigate whether gender differences exist within technology use, since previous research has already demonstrated gender-related biases and inequalities within technological systems, data collection practices, and interaction design (Criado Perez, 2019; Stumpf et al., 2020; Cirillo et al., 2020). Instead, the project investigates how gendered and embodied experiences may help reveal broader normative assumptions surrounding stability, bodily norms, behaviour, cognition, and interaction design itself.

A central argument throughout the thesis is that many contemporary interaction systems implicitly assume a relatively stable, predictable, and standardized user despite human experiences often being contextual, fluctuating, embodied, and socially situated (Bardzell, 2010; Haraway,2006; Tylstedt et al., 2023). Similar critiques have previously been raised within Feminist HCI and critical technology studies, where researchers argue that interaction design frequently normalizes dominant perspectives and interaction patterns while rendering alternative experiences secondary or invisible (Bardzell, 2010; Haraway, 2006). Tylstedt et al. (2023) further argue that technological systems often prioritize standardized efficiency and optimization structures over lived bodily and temporal experiences. From this perspective, women's embodied and cyclical experiences become particularly important within the project not because they represent isolated

"special cases," but because they help expose assumptions that otherwise remain normalized within dominant interaction structures and perceptions of neutrality.

In order to further investigate these challenges and reflections within interaction design, the thesis addresses the following research questions:

Main research question:

How can speculative redesign and reflective interaction design approaches help expose hidden assumptions embedded within contemporary digital systems?

Sub research question one:

How do users perceive neutrality, inclusivity, and adaptation within original and redesigned interaction systems?

Sub research question two:

How might a reflective interaction design toolkit support designers and users in identifying hidden assumptions within digital interfaces?

1.1 Contribution

This thesis contributes to the field of interaction design by exploring how speculative redesign and reflective interaction design approaches may help expose hidden assumptions embedded within contemporary digital systems. Through the development of speculative prototypes and the reflective toolkit Designing Beyond the Default User, the project contributes both practical and reflective perspectives on neutrality, adaptation, inclusivity, transparency, and normative user assumptions within HCI. The thesis further contributes to ongoing discussions within Feminist HCI and adaptive interaction design by highlighting how normalized assumptions surrounding bodies, behaviors, and stability may become embedded and reinforced through everyday interaction systems.

2 Background

This chapter presents theories, concepts, and previous research relevant to gender bias, hidden assumptions, adaptive interaction, and inclusive design within HCI and interaction design. The background further explores how notions of the "default" or "neutral" user influence technological systems, as well as how adaptive and reflective design approaches may support more

transparent and inclusive interaction. Lastly, related design work and existing methodologies connected to feminist HCI, participatory design, and gender-inclusive design practices are presented.

2.1 Gender Bias and Hidden Assumptions in Interaction Design

Criado Perez (2019), in Invisible Women: Exposing Data Bias in a World Designed for Men, discusses how many technologies, products, and systems are developed around male-centered data and assumptions while being presented as gender neutral. Criado Perez (2019) argues that the "default" user in many areas of design, medicine, technology, and infrastructure is implicitly based on male bodies, behaviors, and experiences, creating what she refers to as a gender data gap. This gap becomes visible in both physical and digital design, where women's needs, ergonomics, and experiences are frequently underrepresented or overlooked. One example concerns smartphone design, where increasing phone sizes have been argued to disadvantage many female users due to differences in average hand size and ergonomics. Another example involves voice-recognition systems, where speech recognition software has historically demonstrated higher accuracy for male voices compared to female voices due to male-dominated training datasets.

Further, Criado Perez (2019) highlights how systems often appear "neutral" despite relying on data, prototypes, and design processes that primarily represent male users and perspectives. This creates situations where women more frequently need to adapt themselves to technologies and systems not originally designed around their experiences or bodily conditions. Particular attention is also given to how women's bodily experiences, including hormonal cycles and sex-based medical differences, have historically received less attention within technological and medical research (Criado Perez, 2019).

While Criado Perez (2019) primarily illustrates these issues through real-world examples of data gaps and male-centered design practices, Bardzell (2010) further critiques the notion of universal or neutral design within HCI. Bardzell (2010) argues that all technologies are built upon assumptions regarding their intended users, suggesting that interaction design can never be entirely neutral. This perspective helps frame Perez's (2019) examples not as isolated oversights, but as manifestations of broader normative assumptions embedded within design processes. Similarly, Ehrnberger et al. (2012) argues that technology and design both shape and reproduce societal norms regarding gender, bodies, and behaviour. According to Ehrnberger et al. (2012) assumptions about users become embedded within products and

interfaces during the design process, often remaining invisible until alternative perspectives expose them.

2.2 Who is the “neutral” design really for?

While previous research suggests that gender bias and hidden assumptions frequently exist within interaction design (Criado Perez, 2019; Stumpf et al., 2020; Cirillo et al., 2020), an important question remains: why do many biased systems still appear neutral or universal? Within HCI and interaction design research, several scholars have questioned the assumption that digital systems are truly “gender neutral”. Williams (2014) argues that many technologies are designed and marketed as universal despite often reflecting implicit assumptions regarding a typical or default user. According to Williams (2014), this may result in a form of “gender blindness,” where certain experiences and needs become overlooked while dominant perspectives are normalized as neutral. Similarly, Bardzell (2010) argues that interaction design can never be fully neutral since all systems are shaped by assumptions regarding users, values, priorities, and interaction behaviors. The closer users align with the imagined “ideal user” embedded within a system, the more natural and frictionless the interaction may appear. Consequently, assumptions embedded within interfaces may remain largely invisible to users already represented within the design process itself (Bardzell, 2010).

Stumpf et al. (2020) further argue that interaction design frequently contains hidden gender biases despite claims of universality or inclusivity. Similarly, Both Williams (2014) and Stumpf et al. (2020) further connect these assumptions to the historically male-dominated nature of computing, software development, and engineering fields, which may influence both technological priorities and interaction structures. Stumpf et al. (2020) highlight several examples of how interaction patterns and technological preferences may differ between users, including navigation strategies, privacy concerns, information processing, and approaches to learning new technological systems. For example, women were found to rely more on landmarks and contextual cues when navigating digital environments while also demonstrating higher concern regarding privacy and online trust. Research further suggested that women, on average, displayed lower technological self-efficacy and were less inclined to explore new features through “tinkering,” a playful trial-and-error interaction style often implicitly assumed within software design and interaction systems (Stumpf et al., 2020).

Williams (2014) further argues that interaction systems risk reinforcing hidden assumptions when designers primarily rely on their own experiences, intuitions, and normalized perspectives throughout development processes

rather than continuously reflecting upon diverse lived experiences and user contexts.

Ehrnberger et al. (2012) similarly argue that technology and design both shape and reproduce societal norms regarding gender, bodies, and technological use. According to Ehrnberger et al. (2012), assumptions about users become embedded into products and interfaces throughout the design process, while experiences outside these assumptions risk becoming excluded or overlooked. One example is Volvo's YCC concept car, developed by an all-female design team at Volvo. Although many of the car's features focused on ergonomics, usability, and safety, several were initially dismissed as superficial or "for women." Many of these features later became normalized and integrated into mainstream car design, demonstrating how design solutions addressing overlooked user experiences may initially become perceived as overly specific before later being recognized as broadly beneficial (Ehrnberger, 2012).This perspective closely relates to the concept of the curb-cut effect, which describes how solutions initially developed for marginalized or overlooked user groups frequently benefit a much broader population (Kithmini, 2024). Rather than reducing universality, inclusive and context-sensitive design may therefore contribute to more flexible, supportive, and accessible systems for a wider range of users.

Furthermore, reflections surrounding normalization and perception have also been discussed outside interaction design research itself. Cutler and Scott (1990) demonstrated how conversational participation may be perceived differently depending on what becomes socially normalized within communication patterns. Their findings suggested that asymmetries may remain difficult to identify when dominant behaviors and perspectives become repeatedly framed as balanced, neutral, or universal.

Together, these perspectives highlight how assumptions regarding bodies, cognition, behavior, and technological interaction may become embedded within systems perceived as neutral or universal. Within interaction design research, this has contributed to growing discussions surrounding who digital systems are designed around and how alternative or fluctuating user experiences risk becoming overlooked within dominant interaction structures.

2.3 Adaptive, Context-Aware, and Temporal Interaction

Tylstedt et al. (2023) argue that many contemporary interaction systems continue to be structured around assumptions of stable users, predictable behaviors, and standardized notions of time, productivity, and interaction. Within temporal interaction research, the authors critique how technologies such as self-tracking systems often represent bodily experiences through

fixed and quantifiable structures despite human experiences frequently being contextual, fluctuating, embodied, and socially situated. Similar perspectives have previously been discussed by Haraway (2006), who argues that knowledge, experiences, and bodies are always situated rather than universal or detached from context. Bodily conditions, emotions, motivation, cognitive states, and contextual needs continuously change depending on factors such as stress, health, environment, and social situations. Together, these perspectives advocate for interaction paradigms that better acknowledge uncertainty, changing bodily conditions, and diverse temporal experiences rather than assuming consistent users and routines (Haraway, 2006; Tylstedt et al., 2023).

Adaptive User Interfaces (AUIs) and context-aware systems attempt to respond dynamically to users' changing needs, environments, and behaviors (Dubiel et al., 2022). Rather than presenting identical interfaces to all users, adaptive systems may alter layouts, interaction methods, recommendations, or complexity based on contextual information or user feedback. Previous research has explored adaptive interaction through technologies such as speech recognition, gesture-based interaction, adaptive layouts, and emotion-aware systems (Dubiel et al., 2022). For example, emotion-aware systems such as EmotiControl investigate how interfaces may adapt based on users' detected emotional states through technologies such as facial recognition, behavioral analysis, and physiological sensing (Alipour et al., 2023). Similarly, Dubiel et al. (2022) argue that adaptive systems should not only respond to user behavior itself, but also to contextual and environmental factors such as interruptions, cognitive load, emotional states, and physical surroundings.

At the same time, researchers have also highlighted challenges related to transparency, privacy, agency, and technological control within adaptive interaction. Dubiel et al. (2022) argue that adaptive systems continuously struggle to balance automated adjustment with user autonomy and manual control, while Jacobs and Evers (2023) emphasize the importance of users understanding and influencing how systems interpret their behaviors and contextual experiences. As systems increasingly rely on behavioral interpretation and contextual data collection, questions therefore emerge regarding how users remain aware of the adaptive processes shaping their interaction experiences (Jacobs and Evers, 2023) .

2.4 Empowerment, Agency, and the Risks of Adaptive Systems

Within HCI, adaptive and self-tracking technologies are often associated with empowerment, personalization, and improved decision-making. However, several researchers have questioned whether technological adaptation

automatically results in increased user agency. Tylstedt et al. (2023) critique how self-tracking technologies frequently frame bodily experiences through measurable, efficiency-oriented, and quantifiable technological structures. This risks reducing lived bodily experiences into systems of optimization rather than acknowledging contextual and embodied human experiences (Tylstedt et al., 2023). Jacobs and Evers (2023) similarly discuss ethical concerns surrounding adaptive and self-tracking technologies within femtech applications. The authors specifically highlight risks related to privacy, commercialization of intimate bodily data, and gender stereotyping within adaptive technologies. Rather than rejecting adaptive systems entirely, Jacobs and Evers (2023) instead propose "capability-sensitive design," emphasizing transparency, contextual sensitivity, and users' ability to make autonomous decisions regarding their own bodies and experiences.

Concerns regarding transparency and technological authority have also been discussed within adaptive interaction research. Dubiel et al. (2022) argue that adaptive systems continuously struggle to balance automated adaptation with user autonomy and manual control. Similarly, Bardzell (2010) argues that interaction systems may shape users' identities and behaviors through embedded assumptions and interaction structures. One example lies within EmotiControl that similarly demonstrates how interfaces may continuously interpret users through behavioral sensing, emotional recognition, and contextual monitoring (Alipour et al., 2023). While such systems may support contextual adaptation and accessibility, Alipour et al. (2023) also raise concerns regarding surveillance, bodily autonomy, and user control within adaptive technologies.

Cirillo et al. (2020) further discuss how AI systems may both reinforce and reduce existing sex and gender inequalities depending on how they are designed and trained. The authors emphasize the importance of explainability, visibility of assumptions, and user awareness within adaptive and AI-driven systems (Cirillo et al., 2020).

Together, these perspectives highlight how adaptive interaction design increasingly involves questions related to transparency, agency, contextual understanding, and technological control rather than efficiency and personalization alone.

2.5 Feminist HCI, and Reflective Design

Bardzell (2010) argues that feminist perspectives within HCI challenge assumptions of neutrality and universal design by emphasizing participation, advocacy, self-disclosure, and critical reflection as central design values. According to Bardzell (2010), interaction design can never be fully objective since every design is founded on assumptions about users. Rather than

removing assumptions entirely, feminist HCI instead emphasizes making assumptions more visible and open to critique throughout the design process while acknowledging diverse and situated user experiences.

Reflective and speculative design approaches have further explored how alternative or provocative design concepts may expose hidden assumptions surrounding technology and users (Ehrnberger et al., 2012). The prototypes Drill Dolphia and Mega Hurricane Mixer prototypes, for example, intentionally reversed gender-coded aesthetics and functions in order to challenge assumptions regarding what technologies are perceived as neutral, masculine, or feminine Ehrnberger et al (2012).

Research through Design (RtD) similarly positions design practice itself as a form of knowledge production and critical inquiry rather than solely problem-solving (Zimmerman et al., 2007). Within RtD, speculative prototypes and design artifacts may function as reflective tools capable of provoking discussion, exposing assumptions, and exploring alternative futures or interaction paradigms. Rather than evaluating design only through functionality or efficiency, reflective design approaches therefore emphasize how artifacts may encourage critical reflection surrounding users, values, behaviors, and societal norms (Zimmerman et al., 2007).

Together, these perspectives suggest that interaction design may benefit from more reflective, participatory, and pluralistic approaches capable of exposing assumptions that otherwise remain normalized within design processes. These perspectives form an important theoretical foundation for the reflective toolkit proposed within this thesis.

2.6 Related Work and Existing Design Approaches

Several existing approaches within HCI and interaction design have attempted to address issues related to bias, inclusivity, reflection, and alternative design perspectives within technological systems.

2.6.1 Fairness Testing and Bias Detection

Galhotra et al. (2017) present Themis, a fairness-testing methodology developed to identify discriminatory behavior within software systems. The authors distinguish between necessary computational discrimination, such as categorizing inputs within machine learning systems, and unfair software discrimination caused by biased datasets or problematic causal relationships between inputs and outputs. Themis demonstrates how hidden bias may become embedded within seemingly neutral software systems and highlights the importance of critically evaluating technological decision-making processes (Galhotra et al., 2017).

2.6.2 Gender-Reflective Design Frameworks

Draude and Maaß (2018) propose GERD (Gender-Extended Research and Development), a framework intended to integrate gender perspectives throughout all stages of the design process. GERD encourages designers to critically reflect upon questions such as who benefits from a design, whose needs are prioritized, and who may be excluded through existing assumptions. Rather than treating inclusivity as a final accessibility adjustment, the framework emphasizes continuous reflection throughout development processes. The authors describe GERD as a bridge between technological development and social perspectives on gender and diversity (Draude and Maaß, 2018).

2.6.3 Speculative and Reflective Design Approaches

Drill Dolphia and Mega Hurricane Mixer projects demonstrate how speculative and reflective design approaches may expose hidden assumptions surrounding gender and technology Ehrnberger et a.l (2012). By intentionally reversing gender-coded aesthetics and functions between a drill and a kitchen mixer, the project challenged assumptions regarding which technologies are perceived as neutral, professional, masculine, or feminine (see Figure 1). Rather than functioning as purely practical redesigns, the artifacts primarily aimed to provoke discussion and reflection surrounding normalized technological assumptions (Ehrnberger et al., 2012).



Figure 1 Drill Dolphin and Mega Hurricane Mixer (Ehrnberger et al., 2012).

2.6.4 Exploratory Interface Design Systems

Swearngin et al. (2020) present Scout, a mixed-initiative design system developed to support rapid exploration of alternative interface layouts (see Figure 2). By allowing designers to quickly generate and compare multiple

interface structures, Scout encourages broader exploration of design possibilities during ideation phases. The authors argue that such systems may help designers move beyond conventional or familiar interface solutions by supporting experimentation and alternative interaction structures.

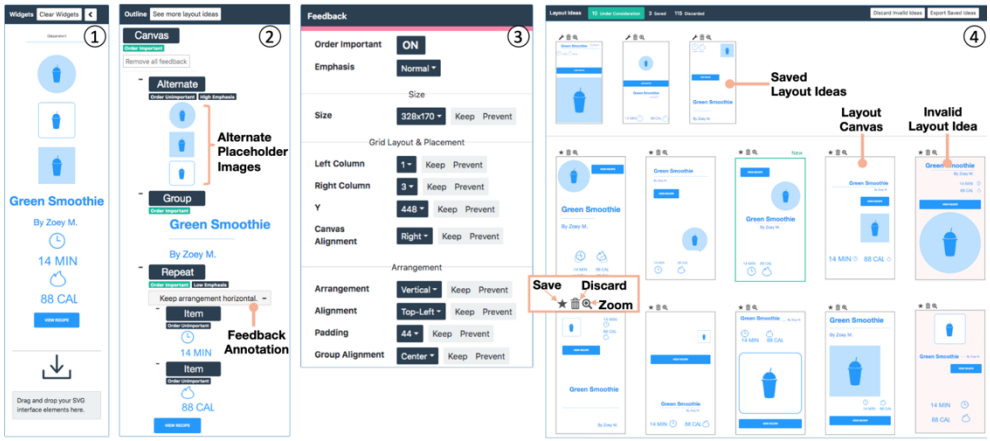


Figure 2 Swearngin et al. (2020)

3 Methodology

This chapter presents the methodological approach and design process used throughout the thesis. The project was conducted through an exploratory and iterative. The overarching methodology consisted of an iterative Research through Design (RtD) process combining speculative redesign, participatory evaluation, reflective analysis, and toolkit development. The aim was to investigate how hidden assumptions emerge within interaction design and how reflective design approaches may support more inclusive and adaptive systems.

3.1 Methodological Framework & Methods

3.1.1 Research Through Design

This thesis was conducted using a Research through Design (RtD) approach. According to Zimmerman et al., (2007), RtD positions design practice itself as a method for generating knowledge within HCI and interaction design research. Rather than viewing prototypes and design artifacts solely as final solutions, RtD treats the design process as a way to explore questions, challenge assumptions, and investigate alternative possibilities through iterative making and reflection. Within RtD, knowledge emerges not only

from analysis and observation, but also through the creation, testing, and evaluation of design artifacts. Zimmerman et al. (2007) further argue that RtD is particularly valuable when addressing complex, subjective, or emerging interaction problems that may be difficult to investigate through traditional scientific methods alone. By designing speculative concepts and interactive prototypes, researchers are able to explore future possibilities, uncover tensions, and generate discussions surrounding technology, users, and interaction practices (Zimmerman et al., 2007).

The RtD approach was considered suitable for this thesis since the project aimed to investigate hidden assumptions within interaction design through speculative redesigns, reflective interaction concepts, and iterative user testing. Rather than attempting to measure a single objective outcome, the project focused on exploring how assumptions become embedded within digital systems and how alternative interaction approaches may influence users' perceptions of neutrality, inclusivity, transparency, and adaptation.

3.1.2 Exploratory Literature Review

An exploratory literature review was conducted throughout the project in order to contextualize the research topic, identify existing perspectives within HCI and interaction design, and support the development of the thesis direction. The literature review helped narrow the scope of the project by identifying recurring themes, tensions, and gaps related to hidden assumptions, gender bias, adaptive interaction, inclusivity, empowerment, and contextual design within digital systems.

Academic sources were primarily collected through Google Scholar and included research from multiple fields connected to the thesis topic, including interaction design, Human-Computer Interaction (HCI), adaptive systems, feminist HCI, participatory design, ethics, and accessibility research. The interdisciplinary approach was intentionally chosen in order to explore both technical and socio-cultural perspectives surrounding interaction design and assumptions embedded within digital systems.

While the majority of the references used within the thesis were academic, certain non-academic sources were also included to broaden the contextual understanding of the topic. One example is Criado Perez's(2019) Invisible Women, which contributed perspectives regarding how hidden assumptions and structural bias may appear within everyday systems and technologies. Although not an academic research publication, such sources provided valuable contextual reflections and real-world examples relevant to the exploratory and reflective nature of the thesis.

The exploratory literature review further helped identify existing claims, frameworks, and research gaps within the field, allowing the project to build upon previous research rather than replicate already established findings.

This process informed both the theoretical framing of the thesis and the development of the speculative redesigns and reflective toolkit.

3.1.3 Interview

As part of the methodological approach, a semi-structured expert interview method was used to gather reflective and experience-based perspectives related to interaction design, adaptive systems, and hidden assumptions within digital technologies. Semi-structured interviews are commonly used within qualitative and exploratory research since they allow participants to elaborate on experiences, reflections, and interpretations through open-ended questioning.

The interview was conducted asynchronously through written e-mail correspondence. This format allowed time for reflection and supported more detailed responses compared to a time-constrained live interview setting. The method was considered suitable for the thesis due to the exploratory and reflective nature of the research topic, where nuanced perspectives and conceptual discussions were prioritized over quantitative data collection.

3.1.4 Focus groups

Focus groups were used as part of the methodological approach in order to gather reflective discussions and collective perspectives related to interaction design, inclusivity, adaptive systems, and hidden assumptions within digital technologies. According to Kitzinger (1995), focus groups are particularly valuable within qualitative research since they allow participants to interact, discuss, question, and reflect upon each other's perspectives rather than responding only individually to the researcher. Through group interaction, participants may reveal assumptions, agreements, tensions, and experiences that might not emerge within isolated interviews. According to Kitzinger (1995), focus groups allow participants to interact, discuss, question, and reflect upon each other's perspectives, making the method particularly useful for exploring collective understandings, social norms, and shared assumptions. This was considered particularly relevant for the present thesis, since the project aimed to explore how users collectively interpret and negotiate assumptions surrounding inclusivity, adaptation, and "neutral" interaction design.

3.1.5 Speculative Redesign & Prototype Development

Speculative redesign and prototype development were used as central methods throughout the thesis in order to explore alternative interaction possibilities and expose hidden assumptions embedded within existing digital systems. According to Dunne and Raby (2013), speculative design can function as a critical and reflective design approach that challenges dominant

norms, assumptions, and expectations surrounding technology and society. Rather than focusing solely on commercial problem-solving or optimization, speculative design aims to encourage discussion, reflection, and alternative ways of imagining interaction futures (Dunne and Raby, 2013).

The method was further inspired by speculative design projects Drill Dolphia and Mega Hurricane Mixer, where redesign and inversion of gender-coded products were used to expose hidden assumptions surrounding gender and technology (Ehrnberger et al, 2012). Rather than aiming to create commercially optimized products, prototypes such as Drill Dolphia and Mega ,redesigns functioned as reflective artifacts that encouraged discussion and critical thinking regarding what types of technologies are perceived as neutral, professional, masculine, or feminine. This perspective strongly influenced the present thesis, where speculative redesign was similarly used as a reflective method for making otherwise normalized assumptions within interaction design more visible in Ehrnberger et al. (2012).

This approach was considered particularly relevant for the present thesis since the project aimed to investigate how assumptions surrounding "neutral" interaction design become visible when alternative and more adaptive interaction approaches are introduced.

3.1.6 User Testing & Participatory Evaluation

User testing and participatory evaluation were used throughout the project in order to gather qualitative reflections regarding the speculative redesigns and adaptive interaction concepts developed within the thesis. Rather than focusing solely on usability metrics or task efficiency, the evaluations emphasized participants' interpretations, reflections, and experiences related to inclusivity, transparency, adaptation, and user agency within digital systems.

Participatory evaluation approaches were used throughout the project in order to allow participants to reflect upon, discuss, and react to speculative design concepts throughout the iterative design process. The method was considered particularly relevant for the present thesis since the project aimed to investigate how assumptions embedded within interaction design become perceived, negotiated, and questioned through alternative interface concepts and reflective interaction approaches.

3.1.7 Data Collection

The project primarily generated qualitative material throughout the iterative RtD process. Data collection consisted of written responses from the expert interview, reflections and discussions from focus groups, observations and qualitative feedback gathered during user testing sessions, as well as notes

and reflections produced throughout the speculative redesign and prototyping process.

Additional material emerged through participatory discussions surrounding the redesigned concepts, where participants reflected upon themes related to inclusivity, transparency, adaptation, neutrality, and user agency within digital systems. Observations regarding participants' reactions, interpretations, and negotiations of the speculative redesigns were documented continuously throughout the evaluation process.

The collected material primarily consisted of qualitative insights, reflections, observations, and discussions rather than quantitative measurements or performance-based usability metrics. This approach aligned with the exploratory and reflective nature of the thesis, where the aim was to investigate how assumptions embedded within interaction design become perceived, interpreted, and questioned through alternative interaction concepts and participatory evaluation.

3.1.8 Thematic Analysis & Affinity Mapping

The collected qualitative material was analyzed using a thematic analysis approach. According to Braun and Clarke (2006), thematic analysis is a method used to identify, analyze, and interpret recurring themes and patterns within qualitative data. Thematic analysis is particularly suitable for exploratory and reflective research since it allows researchers to examine how participants describe experiences, perceptions, tensions, and meanings across different forms of collected material (Braun and Clarke, 2006).

Within the present thesis, thematic analysis was considered appropriate due to the exploratory nature of the project and its focus on assumptions, inclusivity, adaptation, neutrality, transparency, and user agency within interaction design. Rather than focusing on quantitative measurements or statistical patterns, the analysis emphasized recurring reflections, perceptions, and discussions emerging throughout interviews, focus groups, user testing, and participatory evaluation activities.

To support the thematic analysis process, affinity mapping was used as a visual synthesis method for organizing and clustering qualitative insights. According to Krause and Pernice (2024), affinity diagrams are commonly used within UX and interaction design research to identify relationships, recurring observations, and thematic patterns across collected material. Through grouping observations, quotes, reflections, and participant feedback into thematic clusters, affinity mapping may support researchers in identifying broader patterns and connections that might otherwise remain difficult to recognize (Krause and Pernice ,2024)

The combination of thematic analysis and affinity mapping was considered particularly relevant for the thesis since the project aimed to explore complex

and often subjective interaction experiences related to hidden assumptions, contextual adaptation, and perceptions of “neutral” interaction design. The methods therefore supported both analytical interpretation and iterative reflection throughout the Research through Design process.

3.2 Design Process

This section presents the practical design process carried out throughout the thesis. While the project was grounded in a Research through Design approach, the process evolved iteratively through literature exploration, interviews, focus groups, speculative redesign, participatory evaluation, affinity mapping, and toolkit development. Rather than following a strictly linear design process, the project continuously shifted between exploration, reflection, redesign, testing, and synthesis as new assumptions and tensions emerged throughout the work.

The design process evolved iteratively throughout the project rather than through strictly separated phases. Insights generated through literature research, interviews, focus groups, user testing, and participatory evaluation continuously informed subsequent redesigns, reflections, and toolkit development. The process therefore functioned as a cyclical and reflective exploration of assumptions embedded within interaction design.

3.2.1 Initial Exploration & Defining the Problem Space

The project began through an initial exploratory phase focused on identifying tensions and assumptions embedded within contemporary interaction design practices. Early reflections emerged from recurring observations regarding how many digital systems are often presented as “neutral” or universally designed despite frequently being shaped around implicit assumptions concerning users’ bodies, behaviors, needs, routines, and technological experiences. Particular inspiration came from Criado Perez’s (2019) Invisible Women, which highlights how systems, products, and infrastructures are frequently designed around male-centered norms while still being presented as objective or universal. These perspectives contributed to an early interest in how hidden assumptions may similarly become embedded within interaction design and digital interfaces.

During the exploratory phase, literature research was conducted across HCI, interaction design, adaptive systems, feminist HCI, accessibility research, self-tracking technologies, and participatory design. The literature exploration gradually revealed recurring themes related to bodily normativity, perceptions of neutrality, technological authority, adaptive interaction, and the relationship between users and data-driven systems.

Several references further highlighted tensions between personalization and surveillance, as well as between technological efficiency and lived human experiences.

The exploratory phase also involved early reflections surrounding fluctuating user experiences and the limitations of static interaction systems. Existing interfaces often appeared designed around assumptions of stable users with consistent needs, focus, energy, and contexts over time. This gradually shaped the central research interest of the thesis: investigating how hidden assumptions become embedded within interaction design and how alternative, more adaptive and reflective interaction approaches might expose or challenge these assumptions.

The exploratory phase ultimately helped establish the conceptual foundation of the thesis by identifying recurring tensions surrounding neutrality, bodily normativity, transparency, and adaptive interaction within contemporary digital systems. The phase further clarified the importance of investigating not only how interaction systems function technically, but also how hidden assumptions become embedded, normalized, and perceived through everyday interaction. These early insights later informed the direction of the interviews, focus groups, speculative redesigns, and participatory evaluations conducted throughout the project.

3.2.2 Interview with Mexhid Ferati

As part of the exploratory research process, one semi-structured expert interview was conducted with Mexhid Ferati, an Associate Professor within Interaction Design with extensive experience in both academic teaching and professional interaction design practice. The interview aimed to gather reflective insights regarding assumptions within interaction design, adaptive systems, fluctuating user needs, and the relationship between user control and technological adaptation.

The interview was conducted asynchronously through e-mail correspondence. A set of open-ended questions was sent to the participant, who later returned written responses. This format allowed the participant time to reflect on the questions and provide more detailed and carefully considered answers compared to a time-limited live interview setting.

The interview further contributed expert perspectives that helped contextualize and refine the direction of the thesis, particularly regarding adaptive interaction, transparency, contextual design, and assumptions surrounding “default” users within interaction design practices. The participant provided informed consent to both participate in the study and to have his name and responses referenced within the thesis.

The expert interview method was considered suitable for the thesis since the project aimed to explore reflective and conceptual perspectives surrounding

interaction design rather than collect quantitative or statistically representative data. The interview therefore functioned as an exploratory and reflective component within the broader Research through Design process. Here are the questions used in the interview:

In your experience, how do interaction designers or design teams typically imagine the “default” or “typical” user during the design process?

Do you think current interaction design practices tend to assume that users behave consistently over time? If so, how might this influence the systems and interfaces we create?

What challenges or limitations exist when designing for users whose needs, energy, focus, or bodily experiences fluctuate over time?

Many technologies today rely heavily on metrics, tracking, and optimization. In your opinion, what are the strengths and risks of designing systems that quantify human behavior and wellbeing?

Do you believe interaction design can become more adaptive to diverse or changing human experiences without becoming overly invasive or surveillance-oriented? If so, what might that look like in practice?

The interview ultimately contributed to several recurring themes later identified through affinity mapping and thematic synthesis, particularly regarding assumptions surrounding stable users, contextual adaptation, transparency, fluctuating human experiences, and tensions between personalization and surveillance (see Figure 3). These reflections later informed both the speculative redesign concepts and the development of the reflective toolkit.

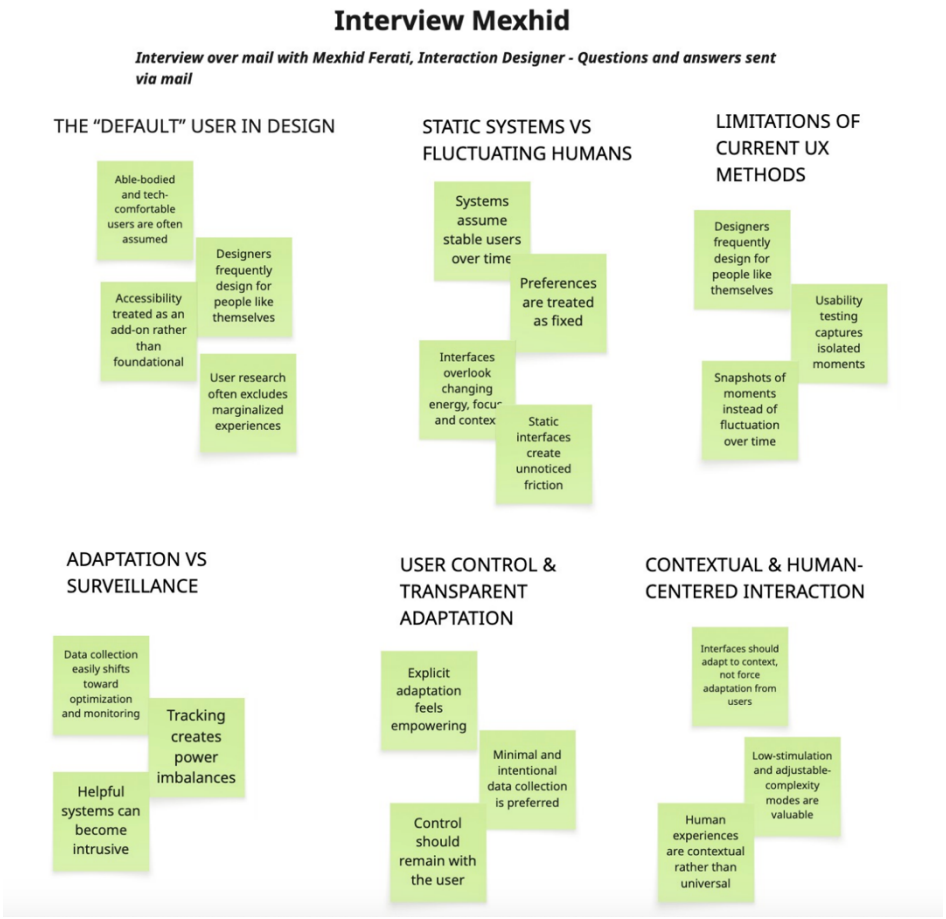


Figure 3 Affinity Mapping of interview

The full interview with Mexhid can be found in Appendix.

3.2.3 Focus groups

Following the initial literature exploration and expert interview, two focus group sessions were conducted in order to further explore perceptions surrounding neutrality, inclusivity, adaptation, and assumptions within interaction design. One focus group consisted of five female participants, while the second consisted of two male participants. The groups were intentionally separated in order to encourage more open discussion and reduce the risk of participants adapting their responses based on mixed-group social dynamics surrounding gendered experiences and technology.

Both focus groups were asked the same set of open-ended questions related to interaction design, adaptive systems, technological assumptions, personalization, bodily experiences, and perceptions of "neutral" digital systems. Rather than functioning as strictly structured interviews, the sessions encouraged participants to reflect collaboratively and build upon each other's responses throughout the discussions. This created a more conversational and reflective environment where participants could

question, expand upon, or relate to each other's experiences and interpretations.

During the later stages of the focus group sessions, examples and findings related to gendered assumptions within everyday products and technologies were introduced as reflective discussion prompts. These included examples discussed in Criado Perez (2019) Invisible Women, such as vehicle safety testing historically based on male crash-test bodies and smartphone dimensions optimized around average male hand sizes. Importantly, these examples were not introduced during the initial stages of the discussions in order to avoid steering participants toward predetermined responses or assumptions. Instead, participants were first encouraged to independently reflect upon experiences and perceptions related to interaction design and inclusivity.

The introduction of these examples appeared to shift the discussions significantly. While several participants initially struggled to identify concrete examples of exclusion or hidden assumptions within technology, later reflections became notably more engaged and emotionally charged once specific cases were revealed. Observations from the sessions suggested that the examples functioned as reflective triggers that helped participants recognize how assumptions may become normalized and difficult to notice within everyday systems and products. These moments later became particularly influential for the direction of the thesis and reinforced the importance of visibility and reflection within the design process.

The focus group discussions ultimately contributed to early thematic clusters related to neutrality, bodily normativity, contextual adaptation, transparency, emotional interaction, and perceptions of inclusivity within digital systems (See Figure 4). These reflections later informed both the speculative redesign concepts and the development of the reflective toolkit.



Figure 4 Affinity Mapping Focus groups

3.2.4 Speculative Redesign & Prototype Iterations

Following the focus groups and early thematic reflections, the project transitioned into a speculative redesign phase focused on exploring alternative interaction approaches and making hidden assumptions within digital systems more visible through design. Rather than designing entirely new systems, the redesign process focused on existing and widely recognized platforms such as ChatGPT and Google Maps. These systems were selected due to their familiarity, broad usage, and perceived neutrality within everyday digital interaction.

3.2.5 Early Concept Exploration & Paper Sketches

The redesign process began through rapid ideation and low-fidelity paper sketches used to explore possible interface structures, adaptive features, and interaction flows. Early concepts focused on transparency, contextual adaptation, fluctuating user experiences, emotional safety, and user control within adaptive systems. The sketching process allowed multiple ideas and interaction possibilities to be explored quickly before transitioning into higher-fidelity prototypes (see Figure 5 and 6).

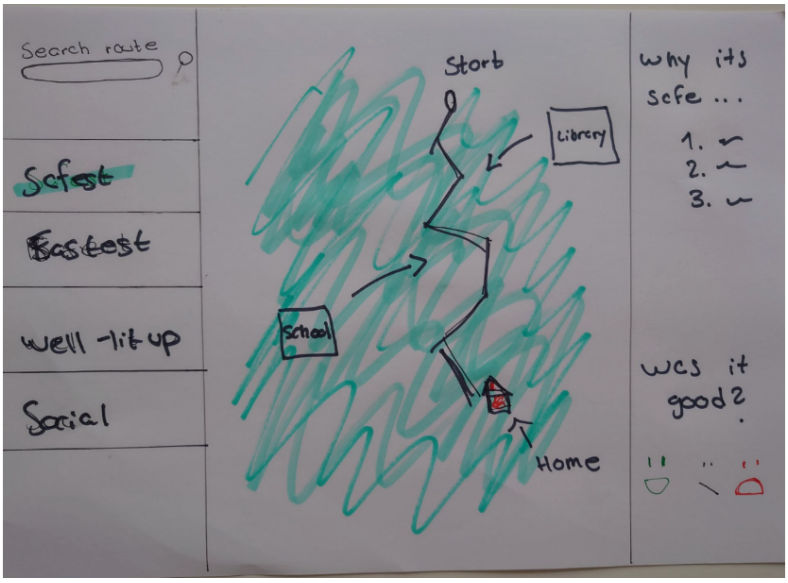
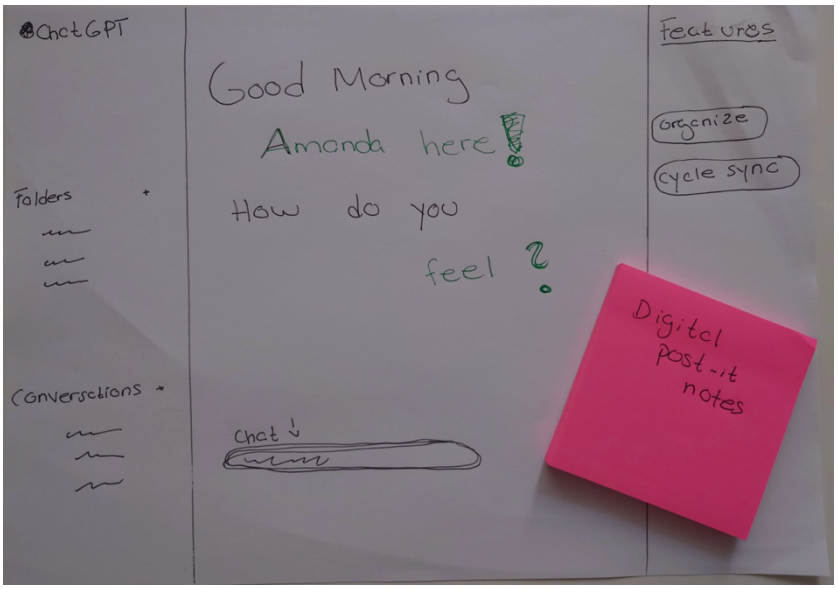


Figure 5 Sketch of Google Maps redesign



Figue 6 Sketch of ChatGPT redesign

3.2.6 AI-Rendered Explorations & Digital Prototyping

Following the initial paper sketches, the redesign concepts gradually transitioned into AI-rendered visual explorations and interactive digital prototypes. Early AI-generated renderings were used to explore visual atmosphere, interface hierarchy, emotional tone, and alternative interaction aesthetics before developing more interactive systems (see Figure 7 and 8).

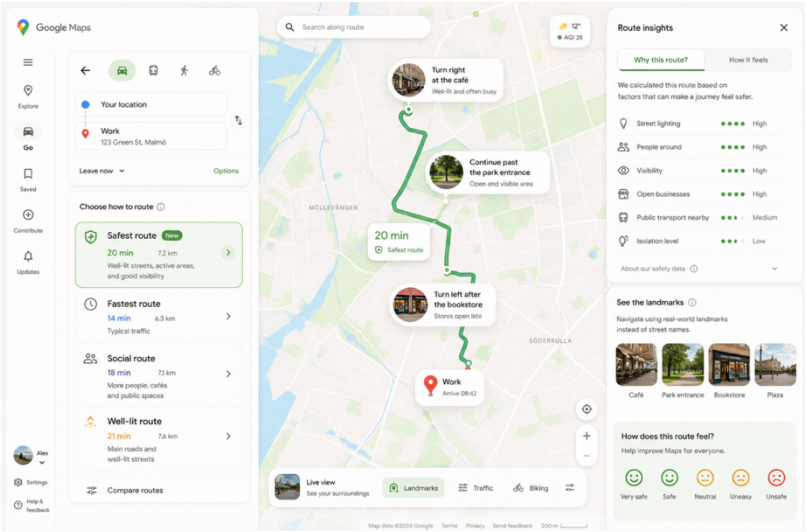


Figure 7. Prototype 1 of google maps

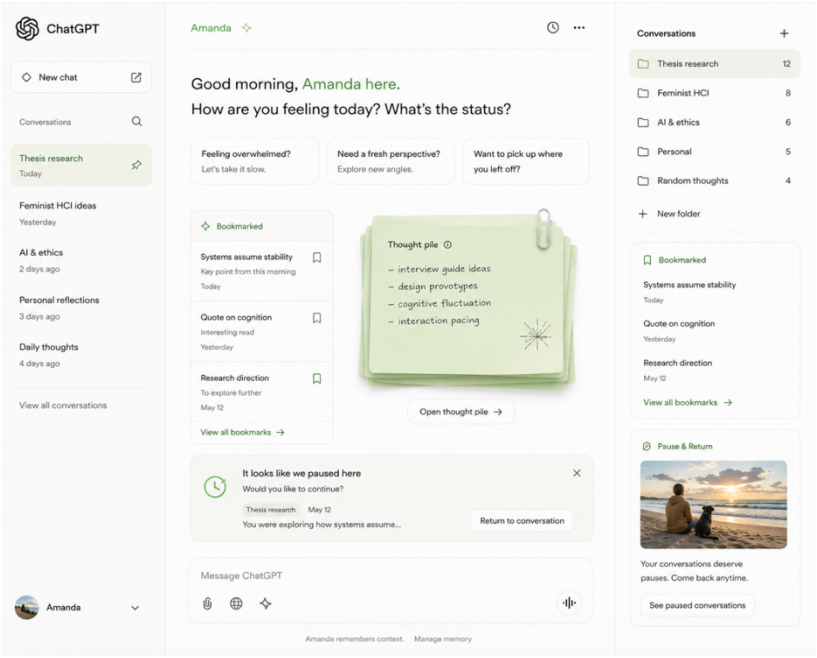


Figure 8 Prototype 1 of ChatGPT

As the project progressed, digital prototypes were iteratively developed using Lovable (Lovable, 2026). The prototypes explored features such as visible contextual adaptation, adjustable interaction modes, transparency surrounding system behavior, and reflective interaction mechanisms intended to expose assumptions embedded within existing interaction patterns. Particular attention was given to designing interfaces that allowed users greater insight and control over how adaptive features operated (see Figure 9 and 10)

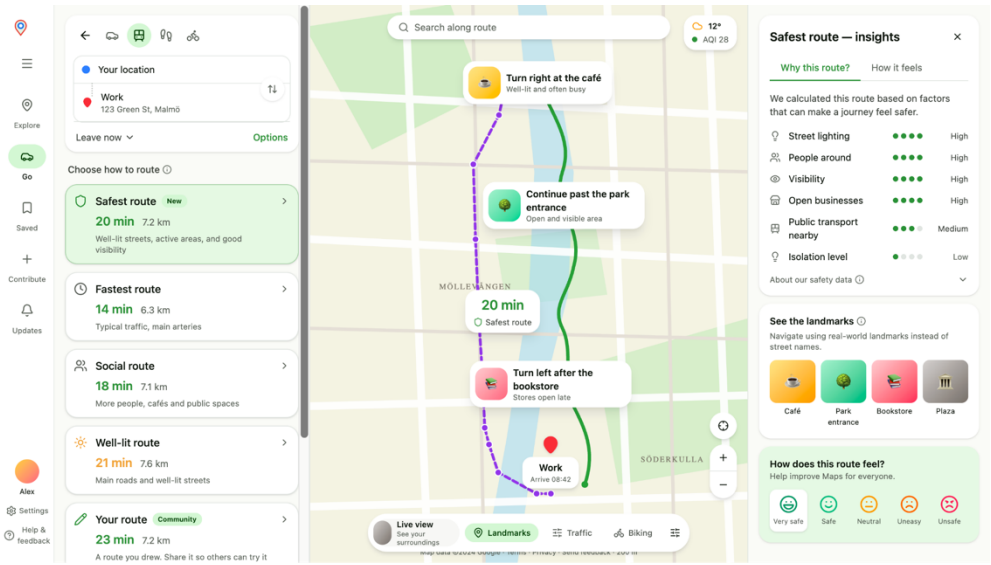


Figure 9. Prototype 2 of google maps

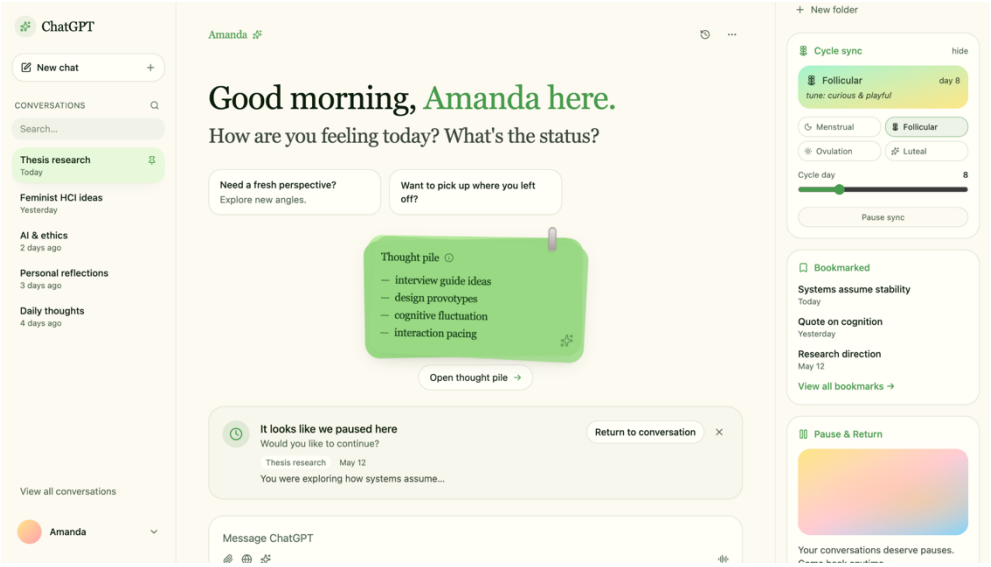


Figure 10 Prototype 2 of ChatGPT

The redesign process evolved iteratively through continuous cycles of prototyping, discussion, user testing, reflection, and redesign. During the first phase of interactive prototyping, four initial user testing sessions were conducted in order to explore how participants perceived and interacted with the redesigned systems. Rather than focusing primarily on traditional usability metrics, these sessions emphasized reflective discussions surrounding inclusivity, adaptation, transparency, emotional interaction, and perceived neutrality within the speculative redesigns.

The testing sessions followed a semi-structured format combining interaction tasks, reflective questioning, and observational evaluation.

Participants were asked to interact with both original systems and redesigned prototypes while reflecting upon themes related to contextual adaptation, transparency, emotional interaction, and hidden assumptions embedded within the interfaces. The sessions combined guided interaction tasks with open-ended reflective discussions, allowing participants to elaborate on frustrations, interpretations, emotional reactions, and assumptions that became visible throughout the interaction process

Several reflections gathered throughout the sessions directly influenced subsequent redesign decisions and prototype iterations. For example, one participant reflected upon the redesigned ChatGPT interface by stating:

“I feel like the original I need to adapt... I need to adapt if it does not understand.”

Another participant described the redesigned Google Maps concept as more supportive due to the inclusion of contextual and safety-oriented route options:

“The safest route... I don’t need to adapt to the route, the route adapts more to the context.”

The sessions also revealed tensions regarding perceptions of neutrality and inclusivity. Several participants initially described existing systems as “for everyone,” while later reflections suggested that many interfaces still appeared optimized around specific assumptions regarding navigation ability, technological confidence, emotional interaction, or contextual stability. One participant reflected upon the redesigned ChatGPT system by stating:

“The old one is more neutral for me. It does not consider me and I like that.”

At the same time, other participants expressed appreciation for increased transparency and contextual adaptation:

“Now I can see why the answer is there and the purpose.”

These reflections and testing sessions informed subsequent redesign decisions and contributed to the emergence of recurring themes related to hidden assumptions, adaptation burdens, contextual interaction, transparency, and perceptions of inclusivity within digital systems.

3.2.7 Emergence & Initial Evaluation of the Reflective Toolkit

As recurring themes and tensions continued to emerge throughout the redesign and testing process, the project gradually shifted from redesigning isolated interface concepts toward developing a broader reflective framework intended to support interaction designers in identifying hidden assumptions within digital systems. Rather than existing as a predefined outcome from the beginning of the project, the toolkit emerged iteratively through insights

gathered from literature research, focus groups, affinity mapping, speculative redesigns, user testing, and participatory evaluation sessions.

Recurring observations related to neutrality, adaptation burdens, transparency, fluctuating user experiences, and contextual interaction gradually evolved into five reflective design principles. These principles aimed to support designers in critically examining who systems are designed around, what assumptions remain invisible, who carries the adaptation burden, how adaptive systems operate, and whether interfaces support fluctuating human experiences.

The development of the toolkit was strongly influenced by earlier user testing sessions. When participants compared original systems such as ChatGPT and Google Maps with speculative redesign concepts, many initially described the original interfaces as “neutral” or “for everyone,” while simultaneously perceiving redesigned versions as “more for women” or “for specific groups.” This became particularly significant since many redesigned features focused on broader concepts such as transparency, contextual adaptation, emotional safety, and user control rather than explicitly gendered functionality.

These observations eventually led to the development of Designing Beyond the Default User: A Reflective Toolkit for Inclusive Interaction Design. The first version initially consisted of AI-rendered conceptual explorations intended to explore layout, visual hierarchy, and reflective direction (see Figure 11).

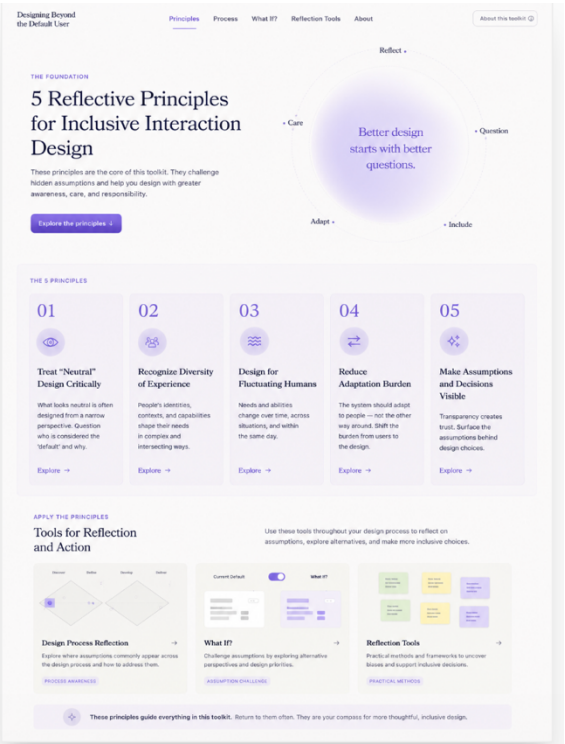


Figure 11. First AI rendering of toolkit

After the initial AI-rendered explorations, the reflective toolkit was further developed into an interactive prototype using Lovable (Lovable, 2026). This first interactive version became the initial toolkit iteration tested with participants in order to evaluate both the reflective framework itself and the usability of the interface structure (see Figure 12).

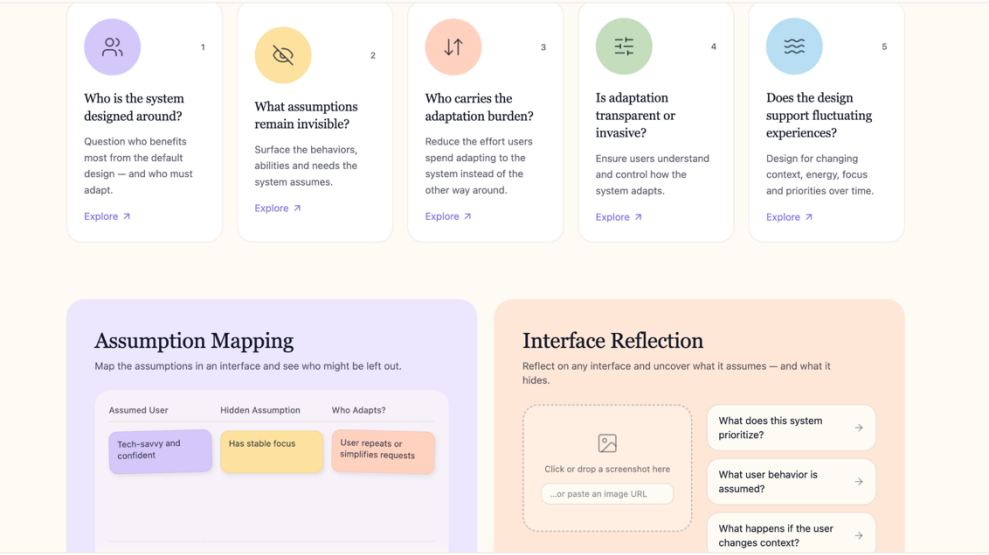


Figure 12 Interface of toolkit.

The first evaluation sessions were conducted with two participants and structured in two separate phases. The first phase focused on evaluating whether the five reflective principles and associated questions supported participants in identifying normalized assumptions embedded within digital systems. Participants were first asked to reflect upon both original interfaces and redesigned prototypes of systems such as ChatGPT and Google Maps without access to the toolkit. Discussions focused on themes related to neutrality, inclusivity, adaptation, transparency, and perceived target users.

Following the baseline reflections, participants were introduced to the reflective toolkit and asked to revisit the same interfaces using the five reflective principles as analytical support.

The five principles focused on:

who the system is designed around,

what assumptions remain invisible,

who carries the adaptation burden,

whether adaptation is transparent or invasive,

and whether the system supports fluctuating human experiences.

The reassessment sessions revealed that participants often began identifying assumptions, priorities, and adaptation burdens that had initially remained

normalized or unnoticed during the baseline discussions. Several participants who initially described existing systems as “neutral” later reflected upon hidden assumptions related to navigation ability, technological confidence, emotional interaction, contextual awareness, and user behavior after using the toolkit prompts.

The second phase of the evaluation focused on the toolkit interface itself, where participants interacted directly with the prototype structure, navigation flow, reflective prompts, and visual hierarchy. While both participants responded positively to the reflective concept and overall direction of the toolkit, several reflections suggested that the interface and information structure occasionally felt overwhelming due to the amount of features, guidance, and reflective content presented simultaneously.

These reflections directly informed the final iteration of the toolkit, where the structure, hierarchy, interaction flow, and visual presentation were further refined in order to improve clarity, accessibility, and usability while still maintaining reflective depth. This final iteration later became the finalized version of Designing Beyond the Default User: A Reflective Toolkit for Inclusive Interaction Design (see Figure 13, 14 and 15).

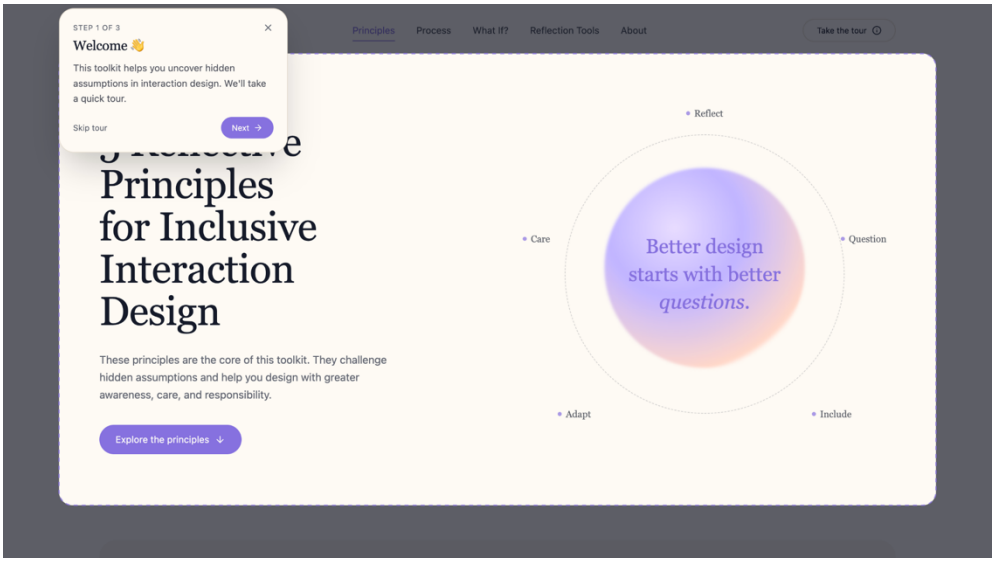


Figure 13. Landing page of the toolkit.

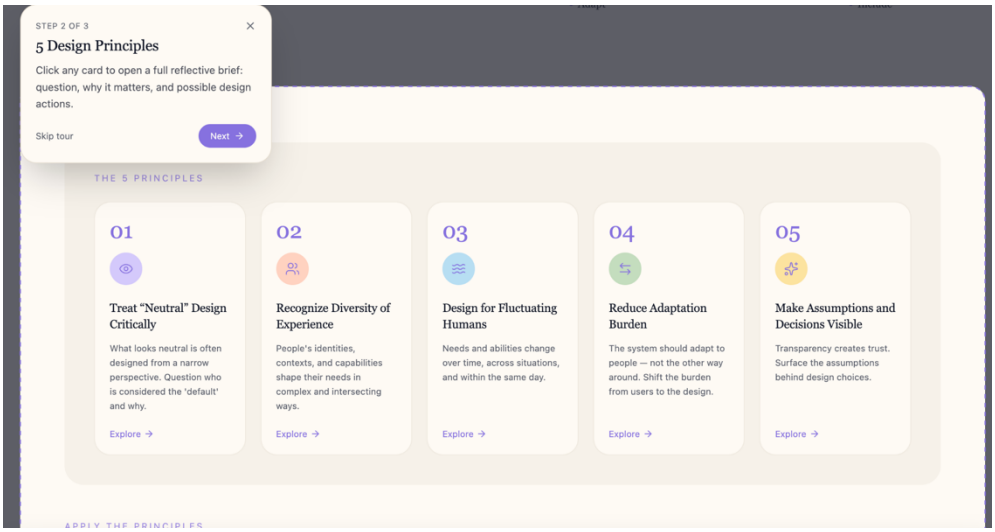


Figure 14. The 5 principles.

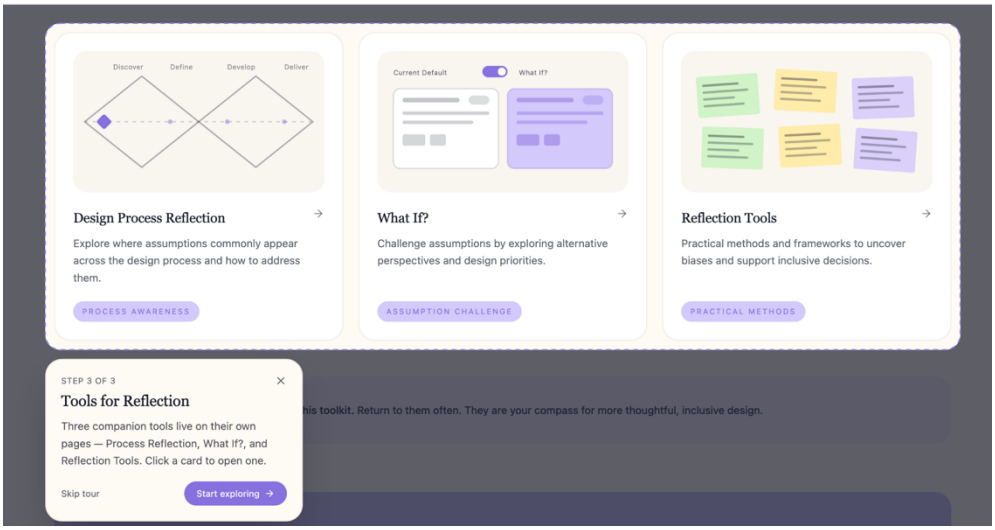


Figure 15. The other helping tools.

3.2.8 Final Iteration & Toolkit Evaluation

The final phase of the project focused on refining and evaluating both the reflective toolkit and the final redesign manifestation developed throughout the Research through Design process. While the earlier speculative redesigns of systems such as ChatGPT and Google Maps primarily functioned as exploratory and reflective vehicles for uncovering invisible default within interaction design, the final phase shifted toward evaluating how the synthesized toolkit functioned in practice and how its principles could be manifested within a more cohesive interaction system.

The final interactive version of Designing Beyond the Default User: A Reflective Toolkit for Inclusive Interaction Design was evaluated through an

additional user testing session conducted with one interaction designer participant. Due to time limitations, the session primarily focused on evaluating the usability, clarity, navigation structure, and reflective depth of the toolkit while also investigating whether the framework supported deeper identification of embedded interaction expectation within digital systems.

The evaluation was structured in multiple stages. The participant was first asked to study the original Google Maps interface while reflecting upon themes related to neutrality, adaptation, transparency, inclusivity, and assumed user behavior. During this initial phase, the participant reflected upon the interface without access to the toolkit in order to establish baseline interpretations and spontaneous reflections surrounding the system.

Following the baseline reflections, the participant revisited the same interface using the reflective toolkit as analytical support. Particular attention was given to the participant's interaction with the five reflective principles, assumption mapping activities, and "What If?" reflection tools integrated throughout the toolkit interface. Unlike earlier testing sessions, the final evaluation focused more heavily on observational analysis rather than direct questioning. Particular attention was therefore given to how the participant navigated through the toolkit, interacted with the reflective tools, and independently engaged with the framework structure throughout the reflective process.

The session suggested that the toolkit supported deeper and more extensive reflective analysis compared to the initial baseline reflections. After interacting with the reflective principles and tools, the participant identified additional assumptions related to navigation ability, contextual awareness, adaptation burdens, and technological expectations that had not initially emerged during the first assessment of the interface. At the same time, the participant still largely described the original Google Maps interface as "neutral," despite identifying multiple embedded assumptions throughout the reflection process. This tension became particularly significant within the broader thesis since it reinforced how deeply normalized certain technological assumptions may remain within everyday interaction systems even when made more visible through reflective analysis.

Reflections gathered throughout the final evaluation sessions directly informed the last redesign iteration of ChatGPT, which functioned as a practical manifestation of the reflective toolkit principles in practice. Rather than continuing to redesign multiple systems simultaneously, the final phase focused specifically on refining the ChatGPT concept through the insights generated from the toolkit evaluation and the broader iterative redesign process.

Several features were further refined based on participant reflections regarding transparency, adaptability, cognitive flexibility, and user control. One of the most appreciated interaction features throughout the testing

process was the visible “context used” functionality, which participants described as increasing both transparency and feelings of control regarding how the AI system generated responses. As a result, the final redesign placed greater emphasis on making contextual reasoning, adaptive behavior, and memory usage more visible and understandable to the user.

Figure 16 show the evolution of the “context used” transparency feature throughout the redesign process. While early redesigns introduced visible contextual memory, the final iteration further clarified how contextual information, adaptive settings, and interaction history influenced AI-generated responses in order to strengthen transparency and user control.



Figure 16 Evolution of the “context used” transparency feature throughout the redesign process.

The final redesign also focused heavily on adaptability and user-controlled interaction. Earlier redesign iterations introduced adaptive and contextual support features, while the final iteration increasingly emphasized customization and flexible interaction structures informed directly by participant feedback regarding overwhelm, rigid interaction patterns, and hidden assumptions. The final interface therefore allowed users to actively add, pause, remove, or modify adaptive features depending on contextual needs and preferences rather than relying on fixed or permanently active system behaviors (see Figure 17).

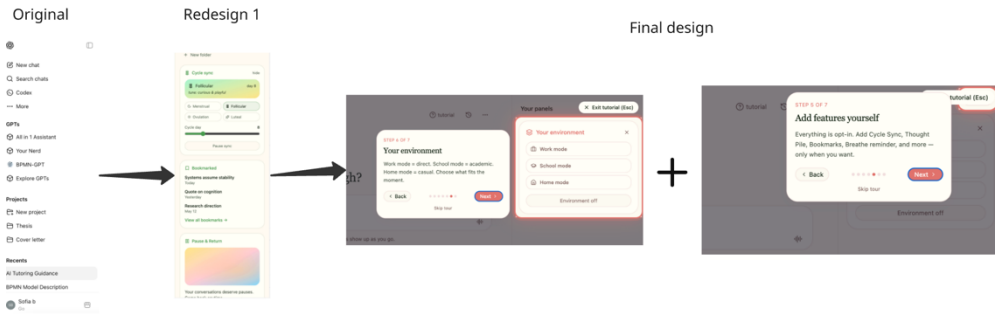


Figure 17 Evolution of the ChatGPT redesign throughout the iterative design process

The final evaluation and redesign phase ultimately demonstrated how the reflective toolkit and speculative redesign process continuously informed one another throughout the project. While the speculative redesigns initially functioned as exploratory vehicles for exposing hidden assumptions within interaction systems, the synthesized toolkit later became a reflective framework used to evaluate, refine, and further develop the final redesign manifestation itself.

The final outcome of the project therefore consisted of both a reflective interaction design toolkit and a speculative redesign manifestation demonstrating how transparency, contextual adaptation, adaptability, and reflective interaction approaches might be integrated into contemporary digital systems. Together, these outcomes functioned as both practical design artifacts and reflective research contributions within the broader Research through Design process.

The final interactive versions of the redesigned ChatGPT concept and the reflective toolkit developed throughout this thesis can be accessed through the following links:

Final ChatGPT Redesign Prototype

Designing Beyond the Default User – Reflective Toolkit

3.3 Summary

To summarize the methodological and design-oriented process of the thesis, the project evolved iteratively through continuous cycles of literature research, participatory discussions, speculative redesign, prototyping, testing, reflection, thematic analysis, and toolkit development. The speculative redesigns of systems such as ChatGPT and Google Maps initially functioned as exploratory vehicles for identifying hidden assumptions embedded within contemporary interaction systems. These recurring

reflections and tensions gradually contributed to the emergence of the reflective toolkit, which later became both a framework for critical reflection and a tool used to evaluate and refine the final redesign manifestation itself. Together, the methods and iterative design process allowed the project to move from exploratory critique toward the development of practical and reflective interaction design approaches aimed at supporting more transparent, adaptive, and critically reflective digital systems.

3.4 Ethical consideration

This thesis followed general principles of good research practice alined with the Swedish Research Council (2025). Since the project involved interviews, focus groups, participatory discussions, and user testing sessions, particular attention was given to informed consent, transparency, anonymity, and the responsible handling of participant data and reflections. Participants were informed about the purpose of the study, how the collected material would be used, and that participation was voluntary throughout the process. Consent forms were distributed prior to participation, and participants were informed of their right to withdraw from the study at any point without explanation. To protect participant privacy, identifying personal information was anonymized and handled confidentially during both data collection and analysis.

Although the study did not involve highly sensitive medical or personal data, the project explored topics related to gender, inclusion, assumptions, and interaction experiences, which may involve personal reflections and subjective perspectives. Discussions surrounding representation, exclusion, and technological experiences can be experienced differently depending on participants' backgrounds and lived experiences. The project further aimed to avoid reinforcing stereotypes or essentialized assumptions regarding gender throughout both the design activities and analysis process. The intention of the redesigned concepts and reflective toolkit was not to design exclusively "for women," but rather to expose and question assumptions embedded within interaction design that may otherwise remain normalized or invisible. Consequently, the project approached gender not as a fixed or universal category, but as one example of how dominant assumptions may shape technological experiences and interaction patterns differently across users and contexts.

Since the thesis specifically investigated hidden assumptions within interaction design, reflexivity was considered particularly important during the interpretation and representation of participant perspectives. Efforts were made to present participant insights responsibly and transparently while acknowledging the exploratory and interpretive nature of the research process.

4 Result

This chapter presents the findings generated throughout the RtD process, including insights gathered from interviews, focus groups, speculative redesign evaluations, user testing sessions, and toolkit evaluations. The results focus particularly on participants' perceptions of neutrality, inclusivity, adaptation, transparency, user agency, and hidden assumptions embedded within contemporary interaction systems. The chapter further presents how recurring reflections and tensions contributed to the emergence of the reflective toolkit and the final redesign manifestations developed throughout the thesis.

Main research question:

How can speculative redesign and reflective interaction design approaches help expose hidden assumptions embedded within contemporary digital systems?

Sub research question one:

How do users perceive neutrality, inclusivity, and adaptation within original and redesigned interaction systems?

Sub research question two:

How might a reflective interaction design toolkit support designers and users in identifying hidden assumptions within digital interfaces?

4.1 Perceptions of Neutrality & Inclusivity

Participants frequently described existing interaction systems such as ChatGPT and Google Maps as "neutral," "normal," or "for everyone" during the initial stages of the focus groups, redesign evaluations, and user testing sessions. Many participants initially struggled to identify assumptions embedded within the systems and instead described the interfaces as broadly universal and equally applicable to all users regardless of context, gender, or bodily experience.

However, as discussions and reflective evaluations progressed, recurring tensions emerged regarding what kinds of users these systems appeared designed around. Participants increasingly reflected upon assumptions related to technological confidence, navigation ability, emotional interaction, contextual stability, bodily norms, and adaptation burdens. Despite these reflections, the original systems were still often largely perceived as neutral even when participants simultaneously identified examples of exclusion,

limitation, or normative assumptions embedded within the interaction structures.

This tension became particularly visible during evaluations of the speculative redesign concepts. Several redesigned features emphasizing transparency, contextual adaptation, emotional support, safety-oriented interaction, and user control were repeatedly described as “more feminine” or “more for women” despite participants simultaneously acknowledging that the features could benefit broader groups of users.

For example, the redesigned Google Maps concept included route alternatives such as “safest route,” intended to support broader contextual awareness and emotional safety during navigation. However, both male and female participants frequently associated this feature specifically with women rather than interpreting it as a generally supportive or adaptive navigation function. At the same time, participants often acknowledged that concerns related to safety, stress, uncertainty, or contextual vulnerability could affect many different users and situations beyond gender alone.

Similar reflections emerged during evaluations of the redesigned ChatGPT concepts. One redesign introduced more adaptive organizational and contextual support features intended to reduce cognitive load and support different working structures, attention levels, and interaction preferences. Rather than primarily reflecting upon these features as supportive for broader cognitive flexibility or contextual adaptation, several participants instead described the redesign as “for people who like organizing their thoughts.” This created an interesting tension where increased adaptability and organizational support became interpreted as specialized preferences rather than questioning whether the original systems themselves might already assume particular working styles, cognitive structures, or interaction behaviours as default.

Another recurring observation emerged when one redesigned ChatGPT concept introduced a single optional feature related to hormonal cycles and contextual wellbeing. Despite the feature being entirely optional and only one small part of the overall system, several participants , both male and female, subsequently described the entire interface as being “more for women.” This reaction became particularly significant within the broader findings since many other redesigned features focused on transparency, contextual adaptation, emotional safety, and user control rather than gender-specific functionality. The reflections therefore suggested that even limited deviation from perceived “neutral” interaction structures could shift how inclusivity and universality were interpreted within digital systems.

Features intended to support adaptability, emotional safety, or contextual awareness were therefore often perceived as less universal than the original systems, even when designed to increase inclusivity across multiple user situations and experiences.

Participants also repeatedly reflected upon how many everyday technologies appeared designed around normative bodily assumptions. Discussions surrounding smartphones, vehicles, navigation systems, and digital interfaces frequently included reflections regarding larger hand sizes, physical strength, right-handedness, good vision, and stable bodily or cognitive abilities. Several participants described these assumptions as deeply normalized within contemporary technological systems and therefore difficult to initially identify without reflective discussion.

The evaluations further suggested that perceptions of neutrality were not necessarily disrupted simply by making assumptions more visible. During the toolkit evaluation sessions, participants often identified additional assumptions and adaptation burdens after interacting with the reflective principles and prompts. At the same time, systems such as Google Maps were still frequently described as “neutral” despite participants simultaneously recognizing hidden priorities, assumed behaviours, and contextual limitations within the interface structures.

The recurring reflections throughout the study therefore suggested that assumptions embedded within interaction systems may remain highly normalized and difficult to identify even when users become more critically aware of them. At the same time, redesign concepts intended to challenge or expose these assumptions often became perceived as more specialized, gendered, or less universally applicable than the original systems they were intended to question.

4.2 Adaptation, Agency & User Control

Recurring reflections throughout the focus groups, interviews, redesign evaluations, and user testing sessions revealed ongoing tensions between technological adaptation, user agency, transparency, and control within digital systems. Participants generally expressed positive attitudes toward adaptive and context-aware interaction systems, particularly when adaptation was perceived as supportive, transparent, and user-controlled. At the same time, several reflections suggested scepticism toward systems that adapted invisibly or relied heavily on hidden tracking, behavioural prediction, or extensive personal data collection.

A recurring observation throughout the evaluations was that participants often described themselves as adapting to systems rather than systems adapting to them. Existing interaction systems were frequently perceived as requiring users to adjust their behaviours, workflows, communication styles, or routines in order to function effectively within the system’s existing structure. This became particularly visible during evaluations of the redesigned ChatGPT concepts, where several participants reflected upon the

relief of not always needing to manually adapt themselves to the system's expectations or limitations.

One participant described the original ChatGPT interaction as requiring continuous adjustment from the user by stating:

"I feel like I need to adapt to original ChatGPT... I need to adapt my prompts if it does not understand."

The reflection suggested that adaptation within existing AI systems was often perceived as dependent upon the user's own ability to intentionally formulate prompts, adjust communication styles, or "learn" how to interact with the system effectively. Several participants appeared to view this type of adaptation as unevenly distributed, since not all users necessarily possess the same technical confidence, prompting strategies, or understanding of how AI systems operate. While systems such as ChatGPT may adapt dynamically to users over time, this adaptation often remained largely invisible and difficult for participants to fully understand or control.

In contrast, the redesigned concepts aimed to shift greater agency and transparency toward the user. Rather than relying primarily on hidden personalization or invisible adaptation processes, the redesigned interfaces allowed users to actively adjust interaction modes, contextual settings, interface structures, and adaptive features while simultaneously receiving visible explanations regarding how and why the system generated responses. Several participants described this increased transparency and controllability as contributing to stronger feelings of trust, understanding, and user agency throughout the interaction process.

Similarly, evaluations of the redesigned Google Maps concepts revealed reflections surrounding contextual adaptation and emotional safety during navigation. Features such as alternative route suggestions based on stress, safety, environmental conditions, or social context were often appreciated when participants felt that the system adapted more clearly to contextual situations rather than forcing users to compensate manually for stressful or unsuitable navigation choices.

One participant reflected upon the redesigned navigation concept by stating:

"The safest route... I don't need to adapt to the route, the route adapts more to the context."

Transparency surrounding adaptive behaviour and data usage also emerged as highly significant throughout the evaluations. Participants repeatedly expressed appreciation for features that made contextual memory, recommendations, or adaptive system behaviour more visible and understandable. In particular, the visible "context used" functionality integrated into the redesigned ChatGPT concepts was frequently highlighted as increasing feelings of trust, control, and transparency regarding how the AI system generated responses.

Several participants expressed discomfort toward adaptive systems when personalization occurred invisibly or without explicit awareness regarding how information was collected or used. While participants were generally open toward intelligent systems, contextual adaptation, and personalized interaction, scepticism frequently emerged when adaptive behaviour became associated with surveillance, intimate bodily tracking, behavioural prediction, or excessive self-quantification. Participants therefore appeared to differentiate between adaptive systems perceived as collaborative and user-controlled versus systems perceived as invasive, overly predictive, or hidden in their operation.

At the same time, participants frequently expressed appreciation for opportunities to customize and adapt interaction systems according to personal preferences, fluctuating needs, or contextual situations. Features allowing users to add, remove, pause, or modify adaptive functionality were generally perceived positively since they supported greater flexibility and individual control over the interaction experience. However, several reflections also suggested that excessive complexity, overwhelming customization structures, or cognitively demanding interaction flows could reduce usability and discourage engagement with adaptive features altogether.

The findings therefore suggested that participants generally valued adaptive and personalized interaction systems when adaptation remained transparent, understandable, and user controlled. At the same time, tensions repeatedly emerged between convenience and surveillance, automation and agency, as well as between supportive adaptation and perceived loss of control within increasingly intelligent and context-aware interaction systems.

4.3 Normative Bodies & Stable User Assumptions

Recurring reflections throughout the interviews, focus groups, redesign evaluations, and user testing sessions suggested that many contemporary interaction systems were perceived as designed around relatively stable, normative, and able-bodied users. Participants repeatedly reflected upon how digital technologies often appeared optimized for users assumed to possess consistent physical, cognitive, emotional, and contextual capacities over time.

Discussions surrounding everyday technologies such as smartphones, vehicles, navigation systems, and digital interfaces frequently included reflections regarding physical strength, larger hand sizes, good vision, right-handedness, technological confidence, and stable bodily or cognitive abilities. Several participants described these assumptions as deeply normalized

within contemporary technological systems and therefore difficult to initially identify without reflective discussion. In particular, references to Criado Perez (2019) Invisible Women frequently triggered stronger participant reactions and reflections surrounding how many everyday products and systems appeared designed around historically male-cantered bodily norms.

Participants also repeatedly reflected upon how interaction systems often appeared designed around assumptions of consistency and predictability in user behaviour. Existing interfaces were frequently perceived as expecting users to maintain relatively stable attention, focus, energy, emotional states, workflows, and contextual situations over time. This reflection became particularly visible throughout both the expert interview and the redesign evaluations, where fluctuating user experiences and contextual variation repeatedly emerged as overlooked within contemporary interaction systems.

The interview with Mexhid Ferati further reinforced these reflections by emphasizing how many interaction systems are built around "a stable version of the user," where systems assume similar levels of focus, energy, context, and interaction patterns across different situations and moments in time. According to the interview reflections, many current evaluation methods similarly prioritize isolated moments of interaction rather than examining how users experience systems across changing bodily, emotional, or contextual states.

Several redesign concepts therefore explored alternative interaction approaches intended to better support fluctuating human experiences rather than assuming stable or universally consistent users. Concepts related to adjustable interaction complexity, contextual adaptation, emotional safety, adaptive support modes, and customizable interaction structures were generally perceived positively when participants felt they supported broader flexibility without becoming invasive or overly predictive.

At the same time, recurring tensions also emerged regarding how much systems should adapt to bodily, emotional, or contextual variation. While participants were generally open toward context-aware and adaptive systems, several reflections suggested discomfort toward technologies that appeared to interpret intimate bodily states, emotions, or behaviours too extensively without explicit transparency or user control. Participants therefore appeared more comfortable with adaptive systems that allowed active user participation and adjustable contextual support rather than systems relying heavily on invisible monitoring or automated behavioural interpretation.

The findings therefore suggested that participants frequently perceived contemporary interaction systems as built around relatively normative and stable assumptions regarding bodies, cognition, behaviour, and interaction patterns. At the same time, the evaluations also indicated growing interest toward more flexible and context-aware interaction approaches capable of

supporting fluctuating human experiences without relying on invasive adaptation or hidden surveillance practices.

4.4 Designer Assumptions & Reflective Interaction Design

Participants frequently reflected upon how many interaction systems appeared designed around assumptions that had become so normalized they were rarely questioned during everyday interaction.

The interview with Mexhid Ferati particularly emphasized how interaction designers and development teams often imagine users based on perspectives and experiences familiar to themselves. According to the interview reflections, many systems appear designed around users who are technologically confident, cognitively stable, able-bodied, and relatively comfortable navigating digital systems without additional contextual support or adaptation. These assumptions were further described as reinforced through existing design practices, user testing structures, and prioritization processes where accessibility and contextual variation may become treated as secondary additions rather than foundational design considerations.

Similar reflections also emerged throughout the focus groups and redesign evaluations, where participants frequently struggled to initially identify assumptions embedded within familiar systems. Several discussions suggested that many interaction patterns and interface structures had become normalized precisely because users continuously adapt themselves to existing systems rather than expecting systems to adapt to broader human variation and contextual complexity.

Throughout the speculative redesign process, recurring tensions repeatedly emerged surrounding who interaction systems are designed around, what forms of behaviour become prioritized as “normal,” and which users are expected to compensate, adapt, or remain invisible within digital interaction structures. The redesign concepts therefore gradually shifted from primarily exploring alternative interface aesthetics toward functioning more as reflective artifacts intended to expose assumptions and provoke discussion regarding what types of interaction behaviours, needs, and experiences become treated as default within contemporary digital systems.

The findings further suggested that reflective interaction design approaches may support designers and users in more actively identifying assumptions that otherwise remain normalized within everyday interaction systems. Several participants described how the redesign concepts and reflective toolkit prompted them to reconsider assumptions they had previously perceived as neutral, natural, or universally applicable. At the same time, the findings also suggested that many assumptions embedded within contemporary technologies remain highly difficult to fully separate from perceptions of neutrality, functionality, and normality.

Recurring reflections throughout the project therefore emphasized the importance of reflection, transparency, and critical questioning within interaction design processes themselves. Rather than framing inclusivity solely as the addition of isolated accessibility features or specialized adaptations, the findings instead suggested the importance of critically examining which users, bodies, behaviours, and experiences interaction systems are implicitly designed around from the very beginning of the design process.

4.5 Reflective Awareness Through the Toolkit

The final evaluation sessions focused specifically on investigating how the reflective toolkit influenced participants' ability to identify and reflect upon hidden assumptions embedded within digital systems. Throughout the evaluations, participants were first asked to assess original interfaces such as Google Maps and ChatGPT without access to the toolkit in order to establish baseline reflections regarding neutrality, inclusivity, adaptation, and user assumptions. Participants were then asked to revisit the same systems using the reflective principles, assumption mapping activities, and "What If?" reflection tools integrated throughout Designing Beyond the Default User: A Reflective Toolkit for Inclusive Interaction Design.

The evaluations suggested that the toolkit supported deeper and more extensive reflective analysis of the interfaces compared to the initial baseline discussions. After interacting with the reflective principles and tools, participants more frequently identified assumptions related to navigation ability, contextual awareness, technological confidence, emotional interaction, adaptation burdens, and fluctuating human experiences that had not initially emerged during the first assessments of the systems.

The participants also appeared to shift focus from evaluating whether systems simply "worked" toward reflecting upon who systems were designed around, who was expected to adapt, and what behaviours or needs became treated as default within the interaction structures. The reflective questions and assumption mapping activities therefore appeared to support broader discussion surrounding hidden priorities, normalized interaction expectations, and invisible adaptation burdens embedded within contemporary digital systems.

At the same time, the evaluations also revealed limitations regarding how easily deeply normalized assumptions become disrupted through reflection alone. Even after participants identified multiple embedded assumptions within systems such as Google Maps, several participants still largely described the interfaces as "neutral" or "for everyone." This recurring tension became particularly significant throughout the thesis since it suggested that

perceptions of neutrality may persist even when assumptions become more visible through reflective analysis.

The toolkit therefore did not necessarily function as a method for definitively "solving" hidden assumptions or bias within interaction design. Instead, the findings suggested that the reflective framework primarily supported greater awareness, discussion, and critical examination surrounding assumptions that might otherwise remain normalized and difficult to articulate within everyday interaction experiences.

Participants generally responded positively toward the reflective direction of the toolkit and appreciated opportunities to critically examine systems beyond traditional usability perspectives alone. Features such as assumption mapping, reflective questioning, contextual prompts, and transparency-focused analysis were frequently described as useful for encouraging broader reflection surrounding interaction systems and design decisions. At the same time, some participants also described earlier toolkit iterations as somewhat overwhelming due to the amount of reflective content, interaction tools, and guidance structures presented simultaneously. This feedback later informed the refinement of the final toolkit structure, navigation hierarchy, and onboarding guidance.

The findings therefore suggested that reflective interaction design tools may support both designers and users in more actively identifying assumptions, adaptation burdens, and normalized interaction expectations embedded within digital systems. At the same time, the evaluations also highlighted how perceptions of neutrality, universality, and technological normality may remain deeply embedded within contemporary interaction practices even when reflective tools make assumptions more visible and discussable.

5 Discussion

5.1 Neutrality, Normalization & the "Default User"

One of the most recurring findings throughout this thesis concerned how existing interaction systems were repeatedly perceived as "neutral" despite participants simultaneously identifying hidden assumptions, adaptation burdens, and exclusionary norms embedded within the interfaces. The findings suggest that neutrality within interaction design may not necessarily represent the absence of assumptions or bias, but rather familiarity with historically normalized design perspectives and interaction patterns.

This became particularly visible during evaluations of the speculative redesign concepts. Redesigned features emphasizing transparency, contextual adaptation, emotional safety, and user control were frequently perceived as “more feminine” or “less universal” than the original systems despite often remaining broadly applicable. Features such as the “safest route” in the redesigned Google Maps concept or optional hormonal-cycle adaptations within the redesigned ChatGPT interface shifted participants’ perceptions of the systems even when the features themselves remained optional or context-dependent. At the same time, the original systems continued to be described as “for everyone” despite participants later identifying assumptions related to technological confidence, bodily norms, and stable user behaviour.

These findings align closely with perspectives presented within Feminist HCI and critical technology studies. Bardzell (2010) argues that interaction design is never neutral since every design process is shaped by assumptions regarding users, values, and interaction behaviours. Similarly, Criado Perez (2019) demonstrates how technologies and systems historically have been designed around male-cantered data and experiences while still being perceived as objective or universal. From this perspective, assumptions may remain difficult to identify precisely because they have become normalized within dominant interaction structures.

An especially interesting observation from the focus groups further reinforced this reflection. Even in situations where male participants objectively occupied more conversational space, several participants still perceived discussions as balanced or even female-dominated. Similar observations were discussed by Cutler and Scott (1990), who demonstrated how conversational participation may be perceived differently depending on what becomes socially normalized within communication patterns. This further supports the notion that what becomes repeatedly normalized may eventually become perceived as neutral or universal even when asymmetries remain present.

The findings further raise questions regarding how interaction systems may reinforce assumptions surrounding users and technological interaction over time. Haraway (2006) argues that technologies should not be understood as neutral tools detached from human experience, but rather as systems actively shaping how humans understand bodies, behaviours, and realities. Similarly, Stumpf et al. (2020) discuss how software systems often implicitly assume exploratory “tinkering” behaviours and technological confidence despite users differing in technological self-efficacy and interaction styles. From the perspective of the present thesis, this raises important questions regarding whether interaction systems designed around assumptions of experimentation and adaptation may unintentionally reinforce exclusion for users who do not already feel represented or comfortable within technological environments.

From my perspective, one of the most striking observations throughout the project was therefore not necessarily that participants identified assumptions within the redesigned systems, but rather how difficult it remained to fully question the neutrality of the original systems themselves.

5.2 Adaptation, Agency & Transparent Personalization

Another important reflection throughout the thesis concerned how adaptive and personalized systems influence users' feelings of agency and control. Within HCI, adaptive technologies are often associated with personalization, empowerment, and improved decision-making. However, the findings of this thesis suggest that adaptation alone does not necessarily increase user agency. Instead, participants' experiences were strongly connected to whether adaptive processes remained transparent, understandable, and open to user control.

This became particularly visible during evaluations of the original ChatGPT interface, where participants described adaptation as dependent upon the user's own ability to formulate prompts correctly or learn how the system "expects" interaction to occur. Adaptation therefore often became experienced as invisible labor placed upon the user rather than collaborative support.

The redesigned concepts instead explored more transparent forms of personalization through features such as visible "context used" explanations, customizable interaction modes, removable adaptive functions, and adjustable interface structures. Participants frequently described these features as increasing feelings of trust, clarity, and control throughout the interaction process.

At the same time, several participants expressed discomfort when systems appeared to interpret emotions, behaviours, or bodily states too extensively without explicit transparency or user control. Adaptation was therefore not necessarily perceived negatively in itself, but rather became problematic when personalization occurred invisibly or when users no longer understood how or why systems adapted to them. This aligns closely with Dubiel et al. (2022), who argue that adaptive systems continuously struggle to balance automated adjustment with user autonomy and manual control. Similarly, Jacobs and Evers (2023) emphasize the importance of technologies supporting users' agency and self-understanding rather than positioning the system itself as the authority on users' needs, bodies, or behaviours.

From my perspective, one of the most important reflections throughout the project was therefore not whether systems should adapt, but rather how adaptation occurs and who controls the process.

5.3 Designing for Fluctuating Human Experiences

Another important reflection throughout the project concerned how many contemporary interaction systems appear designed around assumptions of stable, predictable, and consistent users despite human experiences continuously fluctuating across different contexts and situations. Tylstedt et al. (2023) critique how technological systems often prioritize measurable efficiency, optimization, and standardized temporal structures over lived bodily experiences. Similarly, Mexhid Ferati described how many interaction systems are designed around "a stable version of the user," where systems assume similar needs, energy levels, interaction patterns, and contexts over time (See appendix for full interview).

These reflections became particularly visible throughout the redesign evaluations, where participants frequently appreciated interaction concepts supporting fluctuating experiences such as stress, uncertainty, emotional overload, changing priorities, or contextual vulnerability. The findings therefore suggest a gap between how interaction systems are commonly designed and how human experiences are actually lived.

As Tylstedt et al. (p, 14, 2023) argue, "The technological rationale ends up forcing the soft body into a certain shape". Within the context of this thesis, this became particularly significant since several participants described how they often adapted themselves to technological systems rather than systems adapting meaningfully to contextual human variation.

From my perspective, one of the central risks with contemporary adaptive technologies is therefore not only exclusion, but also the gradual normalization of technological assumptions regarding how humans should behave, regulate emotions, prioritize, and interact with systems. Rather than attempting to perfectly predict or automate human experiences, the redesigned concepts and reflective toolkit instead explored how interaction design might better support fluctuating and contextual human experiences without reducing users to stable behavioural patterns or invisible optimization structures.

5.4 Reflective Interaction Design & the Role of the Toolkit

One of the central outcomes of this thesis was the development of Designing Beyond the Default User, a reflective toolkit intended to support interaction designers in identifying and questioning hidden assumptions embedded within digital systems and design practices. Rather than functioning as a method for automatically "solving" bias or removing assumptions entirely, the toolkit instead aimed to support reflective awareness by making

assumptions, adaptation burdens, and normalized interaction patterns more visible and discussable throughout the design process.

The findings suggest that reflective interaction approaches may support both designers and users in critically examining assumptions that otherwise remain normalized within everyday interaction systems. During the toolkit evaluations, participants frequently identified additional assumptions and excluded perspectives after interacting with the reflective principles and assumption mapping tools, even when they had initially described the systems as “neutral.”

This aligns closely with Bardzell’s (2010) discussion of participation, embodiment, and critical reflection within Feminist HCI, where assumptions become visible through dialogue, situated perspectives, and ongoing critique rather than through supposedly objective or universal design approaches. Similarly, the toolkit shares similarities with approaches such as GERD (Draude & Maaß, 2018), which emphasize the importance of critically questioning who interaction systems are designed around and whose needs risk remaining invisible throughout development processes. Together, these perspectives suggest that assumptions become normalized through interaction systems and perceptions of neutrality, while speculative redesign and reflective interaction approaches may help surface and question these assumptions. At the same time, the evaluations also demonstrated that reflective awareness alone may not fully disrupt deeply normalized perceptions of neutrality within contemporary interaction systems. Although the toolkit frequently supported deeper reflection regarding adaptation burdens, excluded perspectives, and hidden assumptions, participants still often described original systems such as Google Maps and ChatGPT as largely “neutral” even after engaging with the reflective principles.

A similar reflection can be observed within many everyday technologies that continue to appear “neutral” despite being designed around dominant user assumptions. For example, many contemporary smartphones position fingerprint unlock buttons primarily for right-handed interaction. While left-handed users are still able to use these devices, they may nevertheless need to adapt themselves to interaction structures optimized around dominant usage patterns. Such assumptions often remain largely invisible precisely because the majority of users and designers may already align with them. From the perspective of the curb-cut effect (Kithmini, 2024), more adaptable interaction solutions would likely not reduce usability for dominant users but instead increase flexibility and accessibility for a broader range of experiences and interaction styles.

This reflection also relates closely to Ehrnberger et al.’s (2012) speculative design approaches. Through projects such as Drill Dolphia and Mega Hurricane Mixer, Ehrnberger et al. (2012) intentionally exaggerated and reversed gender-coded technological assumptions in ways that made normative structures highly visible and difficult to ignore. Similarly, within

the participatory evaluations conducted throughout this thesis, redesigned systems emphasizing transparency, contextual adaptation, and inclusivity were often perceived as more gendered or targeted toward specific groups despite remaining broadly applicable. From this perspective, deeply normalized interaction assumptions may remain difficult to identify precisely because they already appear universal and familiar.

I argue that one of the most important contributions of the toolkit was therefore not necessarily the specific principles themselves, but rather the shift from attempting to automate inclusivity toward supporting continuous reflection and critical questioning within interaction design practice.

5.5 Methodological Reflections & Limitations

As an exploratory Research through Design project, this thesis should not be understood as producing universally generalizable conclusions regarding interaction design or adaptive systems. Instead, the findings primarily function as reflective and speculative insights generated through iterative redesign processes, participatory discussions, and qualitative evaluations.

One clear limitation concerns the relatively small number of participants involved throughout the focus groups, user testing sessions, and toolkit evaluations. While the qualitative nature of the study allowed for deeper reflections surrounding assumptions, neutrality, adaptation, and inclusivity, the participant pool remains too limited to support broader claims regarding user behaviour or perception.

Another limitation concerns the evaluation of the reflective toolkit itself. While the testing sessions suggested that the toolkit supported deeper reflection regarding hidden assumptions and adaptation burdens, the evaluations remained short-term and exploratory. Future research could therefore investigate how the toolkit functions within longer and more practical interaction design contexts, for example by integrating it into real design projects or educational environments over extended periods of time.

Future work could also further investigate existing reflective design frameworks, adaptive systems, and inclusive interaction approaches already present within both academia and industry in order to more clearly position the toolkit in relation to current design practices and identify remaining gaps.

At the same time, the exploratory nature of Research through Design also functioned as an important strength of the project. Rather than attempting to measure inclusivity or neutrality quantitatively, the project instead aimed to expose, question, and discuss assumptions that often remain normalized and difficult to articulate within everyday interaction system

6 Conclusions

This thesis explored how speculative redesign and reflective interaction design approaches may help expose hidden assumptions embedded within contemporary digital systems. Through iterative redesigns, participatory evaluations, and the development of the reflective toolkit Designing Beyond the Default User, the project investigated how assumptions surrounding neutrality, adaptation, inclusivity, transparency, and user behaviour become normalized within interaction design. The findings suggest that many systems perceived as “neutral” still reflect dominant assumptions regarding bodies, behaviours, and interaction patterns, while more inclusive or adaptive alternatives often become perceived as specialized or gendered. At the same time, the study highlighted how users generally appreciated adaptive technologies when personalization remained transparent, understandable, and user-controlled rather than invisible or overly assumptive. Rather than attempting to remove assumptions entirely, the thesis therefore argues for more reflective and critically aware interaction design practices capable of making assumptions more visible, negotiable, and discussable throughout the design process itself.

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8.1 Interview

1. In your experience, how do interaction designers or design teams typically imagine the "default" or "typical" user during the design process?

In my experience, designers tend to picture someone who is able-bodied, comfortable with technology, and without any specific constraints. This happens partly because of who ends up in user research, and partly because designers often design for people like themselves. Personas can help, but they can also reinforce the same assumptions if they're not built carefully. Accessibility is usually treated as something you add toward the end of the process, which already tells you a lot about who the "default" user is in most projects.

2. Do you think current interaction design practices tend to assume that users behave consistently over time? If so, how might this influence the systems and interfaces we create?

Yes, I think they do. Most systems are designed around a stable version of the user, which means, same focus, same energy, same context every time. You set your preferences once and the system just keeps them. But people are not like that. The same person can have very different experiences from one day to the next, and unfortunately most interfaces are not built with that in mind. It creates friction that is easy to overlook if you are not looking for it.

3. What challenges or limitations exist when designing for users whose needs, energy, focus, or bodily experiences fluctuate over time?

The main challenge is that most of our evaluation methods give us a picture of one moment, not how a person experiences a system over time and across different states. And trying to capture that variation often means more data collection, which puts more burden on the user. In practice, features that would really help people with variable needs, like adjustable complexity or low-stimulation modes, often get cut because they do not show up as priorities in standard usability testing or in client requirements.

4. Many technologies today rely heavily on metrics, tracking, and optimization. In your opinion, what are the strengths and risks of designing systems that quantify human behavior and wellbeing?

When users have real control over their own data, quantification can be useful, because it can support self-awareness and help people make better decisions. The risk is that data rarely stays with the user. Once something is being tracked, there is pressure to use it for other purposes, such as monitoring, optimization, targeting. Even when the original intention is good, the system can easily shift toward serving other interests. So the question is not really whether to measure, but who controls the data and what they are allowed to do with it.

5. Do you believe interaction design can become more adaptive to diverse or changing human experiences without becoming overly invasive or surveillance-oriented? If so, what might that look like in practice?

I think it can, but it requires making choices that go against a lot of current trends. There is a real difference between a system that adapts because you explicitly asked it to, and one that adapts because it has been observing your behavior. The first feels like control; the second can feel intrusive even when it is technically helpful. More adaptive design could mean giving users clear options to switch between, being transparent about what the system has learned, and collecting only what is actually needed. It should be less about the technology itself and more about where the control sits.